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Climate change impacts on crop yields across temperature rise thresholds and climate zones

Bao-Linh Tran, Wei-Chun Tseng & Chi-Chung Chen✉

This study quantifies the projected impacts of climate change on crop yields across temperature rise regimes and climatic zones, using the latest global dataset of site-level process-model simulations of crop responses to climate scenarios. We employed a threshold regression technique to identify and estimate temperature change thresholds and used linear mixed-effects models to assess the climate impacts on crop yields across different levels of temperature rise. The results indicated that warmer temperatures are detrimental to crop yields across countries, with negative impacts exacerbated when temperature increase exceeds threshold values. For instance, for wheat, a 1 °C temperature increase would result in a 6.1% yield loss when the temperature rise is below 2.38 °C; however, when it exceeds 2.38 °C, yield loss would rise to 8.2% per 1 °C warming. Similarly, the loss in rice yields for each °C increase in temperature would increase from 1.1 to 7.1% per °C when the temperature rise surpasses the 3.13 °C threshold. For maize, no threshold effect is found; instead, temperature increase would reduce yields by an average of 4.03% per °C. We also conducted impact assessments by climate zone, categorizing studied sites according to the *Köppen* climate classification system. We found that crop yields in arid regions are most adversely affected by global warming compared to other zones, while adaptive potential is higher for rice and wheat in temperate zones and for maize in continental zones. This study highlights the existence of threshold effects of temperature rise on crop yields and the varying yield impacts among climate zones, informing effective adaptation strategies to enhance global food security.

Keywords Climate change, Crop yield, Temperature thresholds, Mixed-effects model, Meta-analysis, Global food security

Global warming has sped up at an unprecedented rate in recent years. The global average temperature from June 2023 to May 2024 reached a record-breaking 1.63 °C above the 1850–1900 baseline[1]. Experts predict that within the next five years (2024–2028), the temperature will surpass 1.5 °C above the pre-industrial average[2]. Agriculture is the most vulnerable sector facing climate change since crop growth is highly susceptible to changes in weather conditions. In recent years, mean annual temperatures over areas where major crops are cultivated are observed and predicted to increase, causing severe impacts on food production[3]. By disrupting crop production, climate change is undermining food security, especially when global food demand is projected to increase by at least 50% by 2050[4]. To improve food security, the impacts of climate change on crop yields and the effectiveness of adaptation measures under different warming scenarios have been widely studied over the past decades.

At regional scale, literature has shown mixed results in the direction of crop responses to climate change. For instance, the projected impacts of future climate change on wheat yields are found to vary from negative[5–7] to positive[8–11] across studied locations. Similarly, inconsistent trend in climate impacts on maize yields are also predicted[3,12,13]. Ray, et al.[14] investigated how recent climate change affected major global crops and found that climate impacts are negative in Europe, Australia, Southern Africa; positive for countries in Latin America, and mixed for Asia and North and Central America. Besides sites, these discrepancies may also come from the differences in methodology adopted, climate data derived, and management strategies considered[15,16]. Regarding the magnitude of the climate impacts, comparisons showed that different methods mostly generated similar results for the same crop at the country scale, but estimates varied between countries[3,17,18], while the projected impacts of climate change on global crop yields are still largely inconclusive[19].

Compared to using one single crop model, estimating crop responses to future climate change based on an ensemble of extensive published simulations would be more reliable as it sufficiently captures all possible relevant

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processes[20]. Thus, a few meta-analyses have been conducted to provide more consistent impact estimates at the global level[3,21,22]. Challinor, et al.[21] developed a dataset of more than 1,700 published simulations and their results indicated that the average yield of C3 and C4 crops would decrease by 4.9% in response to a 1 °C increase in global mean temperature. In addition, crop yields are also found to increase by 0.53, 0.06, and 7.16% in response to a 1% increase in precipitation, a 1 ppm increase in CO2 concentration, and with the presence of adaptation measures, respectively. Another meta-analysis by Wilcox and Makowski[23] quantified more than 3600 simulations from 90 studies adopting computer and found that wheat production is estimated to decrease by 3.28% per °C of warming, with the impact largely varied among different sites. Zhao, et al.[3] compiled published outcomes estimated by four different methods to project the global impacts of warmer temperatures on crop yields. The study confirmed the negative effects of warming, with yield loss of 7.4, 6, 3.2, and 3.1% per °C for maize, wheat, rice, and soybean, respectively. A meta-analysis Makowski, et al.[22] synthesized simulations from experimental and modeling studies and confirmed the negative impacts of global warmings on crop yields, where the yield change ranges from −2.4% (for C3 crops) to −4.52% (for maize, C4) for an increase of 1 °C. Recently, Abramoff, et al.[16] employed statistical and machine learning models on the Hasegawa, et al.[24]'s latest global dataset of 8703 simulations built on 202 studies published in 1984–2020 and found that 1 °C increase in global mean temperature (RCP4.5-2060, without adaptation) would result in a reduction of maize, wheat, soybean, and rice by 10, 6.5, 5.4, and 2.8%, respectively.

Though providing an overview of the warming effects on global crop yields, the temperature-yield relationship in meta-analysis could still be improved from several points. First, the rising temperatures in reality could not always have constant impacts on crop yields, thus, averaging temperature effect will underestimate the true response of crops to temperature, especially at extremely warm temperatures[25,26]. At regional-scale studies, temperature thresholds for crop growth are determined at 29 °C for maize and 30 °C for soybean in the US[26], 30 °C for each of growing degree days for maize in Africa[27], or wheat in China[28], where crop yields are found to gradually increase or not respond vividly to rising temperatures, but significantly reduce when the temperature crosses the thresholds. As attempts to identify critical thresholds of high temperatures on crop yields are of essential for supporting policy decisions regarding agricultural adaptation measures[29,30], assessing nonlinear temperature impacts on crop production in simulations are suggested[31,32]. In addition, Threshold in temperature change is another useful concept. It was first discussed in the late 1980s when the UNFCCC suggested a 2 °C warming above pre-industrial level as a critical limit for dangerous anthropogenic interference (DAI)[33], and later extended to a broader concept, encompassing various temperature thresholds across sectors impacted by climate change[34,35]. In agriculture, identifying thresholds of temperature change is crucial for informing agricultural policy discussions on appropriate mitigation targets at different regional warming levels[29,36]. However, most studies estimating the climate impacts on crop yields identified thresholds of temperature change by selecting specific warming levels based on expert judgment[21,22,37,38] or data visualization[23,29], rather than quantitative assessments. Therefore, it is important to have thresholds of temperature increase estimated endogenously to investigate the threshold impacts of climate change on specific crop yields. This is one of the major contributions of this study.

The second aspect that could be improved is disaggregating crop responses to warming temperatures and adaptation strategies by climatic zones. In prior studies, locations are usually classified by country and/or continent, and the climate regions are generally defined by latitudes[23,36,39,40]. However, latitude alone could not comprehensively capture the climate condition of a region, since areas located in the same latitude can vary drastically in climate patterns due to the difference in elevation and proximity to large bodies of water[41]. To improve defining climate zones for better assessing regional climate impacts on crop yields, we categorize climate zones following the *Köppen* climate classification system[42], which builds on empirical observations of monthly temperature and precipitation, concerning local vegetation of each region[43]. This is another research contribution of our study.

The main objective of this work is to provide a quantitative synthesis of the climate change impacts and adaptive potential for crop yields across temperature change regimes and climatic zones. To achieve this, threshold regression technique is adopted to determine whether temperature rise thresholds exist in the relationship between temperature change and yield change. We then employed mixed-effects regression models to estimate the impacts of climate change and adaptation strategies on crop yields across different temperature increase ranges. Finally, we quantified the impacts of climate change on crop yields, disaggregated by four distinct climatic zones.

This paper is organized as follows. Sect. "Introduction" discusses the research motivations and objectives. Dataset and methodology are described in Sect. "Materials and methods". Sect. "Results" provides empirical results. Study findings, policy implications, and limitation of the study are discussed in Sect. "Discussion". Sect. "Conclusion" concludes the paper.

Materials and methods

To estimate crop response to climate change across different temperature regimes and climate zones, Hasegawa, et al.[24]'s global dataset is used and threshold regression technique and mixed-effects models are applied.

Global dataset of projected crop response to climate change

The global dataset provided by Hasegawa, et al.[24] builds on 8703 crop model simulations (from 202 studies published between 1984 and 2020) of crop responses to different warming scenarios. The datasets were acquired from two primary sources. The first source is the Aggarwal, et al.[44]'s meta-analysis which includes 99 studies published in 1984–2016, building on the dataset used for the 5th IPCC assessment report (AR5)[21,37]. The second source is the new literature covering the 2014–2020 period from studies considered in the 6th IPCC assessment (AR6). Major process-based crop models, which are commonly used in these AgMIP and IPCC-related studies,

are DSSAT (Decision Support System for Agrotechnology Transfer), APSIM (Agricultural Production Systems sIMulator), EPIC (Environmental Policy Integrated Climate), WOFOST (World Food Studies), and AquaCrop (FAO model for water-driven crop productivity), or multi-model ensembles, facilitating detailed simulations of crop growth under climate change.

To quantify the projected effects of climate change on crop yields, we selected the following relevant variables from the dataset: relative yield change (%), local temperature change (°C), local precipitation change (mm), CO_2 concentration (ppm), and the presence/absence (1/0) of adaptation strategies, reference number, sites (country), and coordinates of sites (latitude, longitude). Precipitation change is converted from into percentage (%) by multiplying with 100 and divided by current annual precipitation (2001–2010) from the 0.5°-grid data (area-weighted average). To ensure the interpretability of the metadata and eliminate potential bias introduced, baseline-corrected data relative to 2005 are used for estimation.

Descriptive statistics of variables are presented in Table 1.

Among major crops, maize has the greatest projected global yield reduction (−8.1%), with significant site-level variation across sites, followed by wheat (−6.2%) and rice (−3.5%), consistent with literature identifying maize as the most heat-sensitive crop. High climate variability is observed in wheat-growing areas, with the lowest base temperature (15℃) and precipitation (614 mm), yet wide ranges in temperature change (mean: + 1.88℃, range: −0.14℃ to 5.55℃) and precipitation change (mean: + 8.38%, range: −677% to + 1470%). This suggests wheat-growing sites will likely face the most significant climate change, posing greater risks and adaptation challenges. Finally, projected adaptation rates are highest for maize (77%), and 46% for both wheat and rice.

Köppen climate classification system

To estimate the impacts of climate change on crop yields in different climate zones, studied regions are categorized following the Köppen-Geiger climate classification system. Based on the idea of defining climate zones by native vegetation, the system was first developed in the late nineteenth century by botanist-climatologist Wladimir Köppen, and later enhanced and modified with collaboration by climatologist Rudolf Geiger. The system categorizes world into five climate zones, based on temperature and precipitation criteria of a region. The five major climate zones are: A-tropical zone (tropical climates), B-arid zone (dry climates), C-temperate zone (humid subtropical climates), D-continental zone (humid continental climates), E-polar zone (polar/alpine climates).

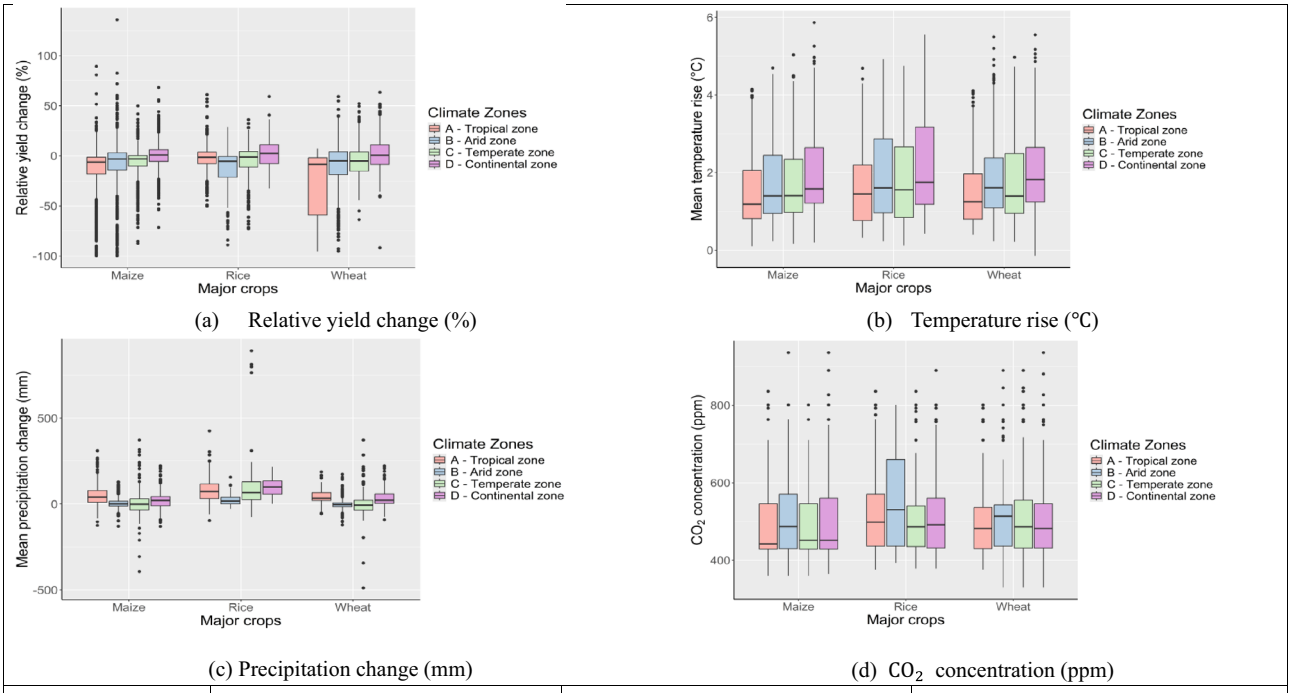
In this study, latitude and longitude coordinates of sites are used to computing climate zones, based on the updated version of the Köppen-Geiger classification system in 2018[42]. Since number of sites located in polar zone in the data set is trivial, the analysis is performed for four climatic zones A, B, C, D. Statistics of variables projected yield change, temperature change, precipitation change, and CO_2 concentration for each crop in four distinct climate zones are illustrated in Fig. 1. The below part is summary of average baseline temperature, baseline precipitation, and adaptation presence rate. Projected yield changes by temperature rise scenarios with and without adaptation strategies are shown in Fig. 2.

For maize, without adaptation, yields consistently decline across all four major climate zones as temperature increases. Notably, yield losses are pronounced when mean temperatures exceed 4 °C. Conversely, implementating adaptation measures mitigates this decline, resulting in a less severe yield reduction. Regarding wheat, there is a slight fluctuation in yield. However, wheat yield losses in tropical zones and arid zones are substantial under both scenarios, with and without adaptation, when temperatures exceed 4 °C. For rice, yields decline across all climate zones without adaptation, with significant reductions observed particularly when temperatures rise above 4 °C. Adaptation are found to significantly stabilize yields for all three crops.

As an aside, adaptation potential (%), which defined as the difference between yield impacts with and without adaptation, disaggregated by different adaptation options are also displayed in Fig. 3.

Variables	Relative change in yields (%)	Temperature change (°C)	Precipitation change (%)	CO_2 concentration (ppm)	Presence of adaptation (1/0)
Maize (4526 obs.)					
Mean	−8.06	1.79	1.56	509.21	0.77
Std. Dev	22.52	1.08	10	111.2	0.42
Min	−100	0.1	−292.32	360	0
Max	135.91	5.87	44.41	935.87	1
Wheat (2212 obs.)					
Mean	−6.18	1.88	8.38	516.63	0.46
Std. Dev	20.93	1.05	179.79	117.37	0.5
Min	−95.51	−0.14	−676.69	330	0
Max	63.59	5.55	4170.24	935.87	1
Rice (1463 obs.)					
Mean	−3.48	1.81	5.6	530.75	0.46
Std. Dev	16.11	1.17	35.68	119.67	0.5
Min	−88.91	0.12	−676.69	376	0
Max	61.13	5.55	542.31	889.98	1

Table 1. Descriptive statistics of variables.



Crop	Wheat				Maize				Rice			
Climatic zones	A	B	C	D	A	B	C	D	A	B	C	D
Base temperature (℃)	25.01	16.44	15.33	9.82	25.04	18.43	15.15	9.28	27.63	18.72	18.52	11.27
Base precipitation (mm)	1170	424.41	788.13	676.81	1438.66	607.63	1158.01	820.94	1590.54	465.34	1490.95	1382.62
Adaptation (1/0)	0.49	0.51	0.37	0.44	0.83	0.78	0.69	0.74	0.53	0.5	0.32	0.39
No. of observations	163	1055	364	630	1392	1249	667	1188	712	199	364	138

Fig. 1. Visualizations and descriptive statistics of variables for each crop by climate zones.

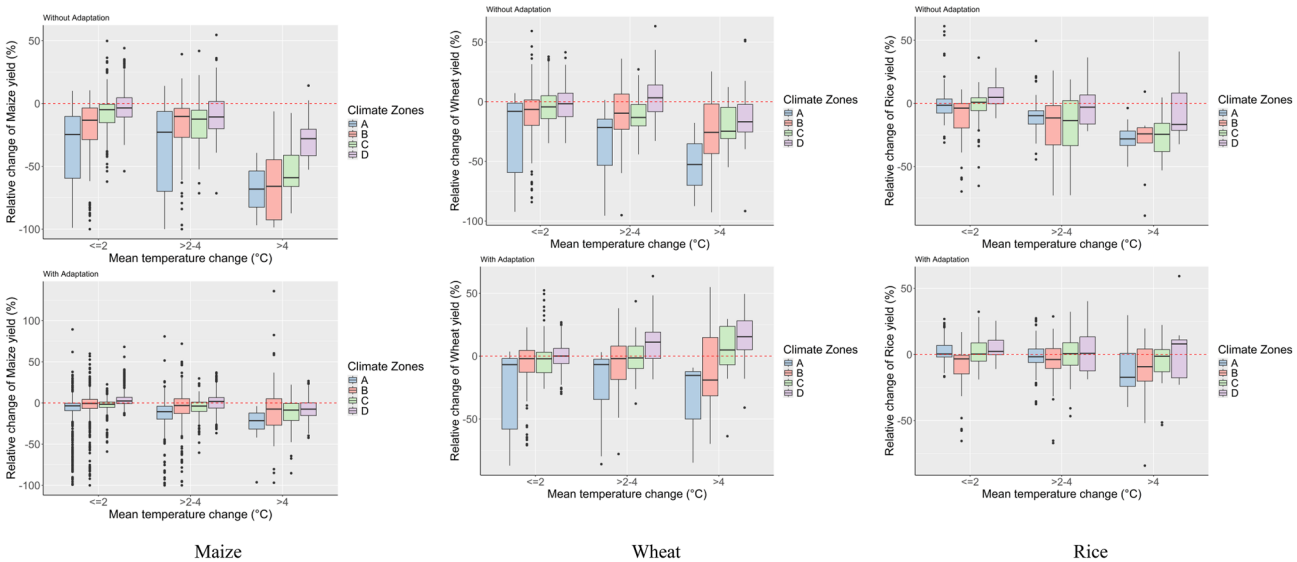


Fig. 2. Projected yield changes by different temperature rise levels, with and without adaptation practices.

Assessing the impacts of climate change and adaptation on global crop yields

We synthesize the projected effects of climate change and adaptation on global crop yields following Wilcox and Makowski[23] and Abramoff, et al.[16]'s model specification. In particular, relative yield change (RCY) is a function of climate factors including temperature change (ΔT), precipitation change (ΔP), CO_2 concentration, and interactions between climatic factors, quadratic terms of climatic factors, and the presence/absence of adaptation.

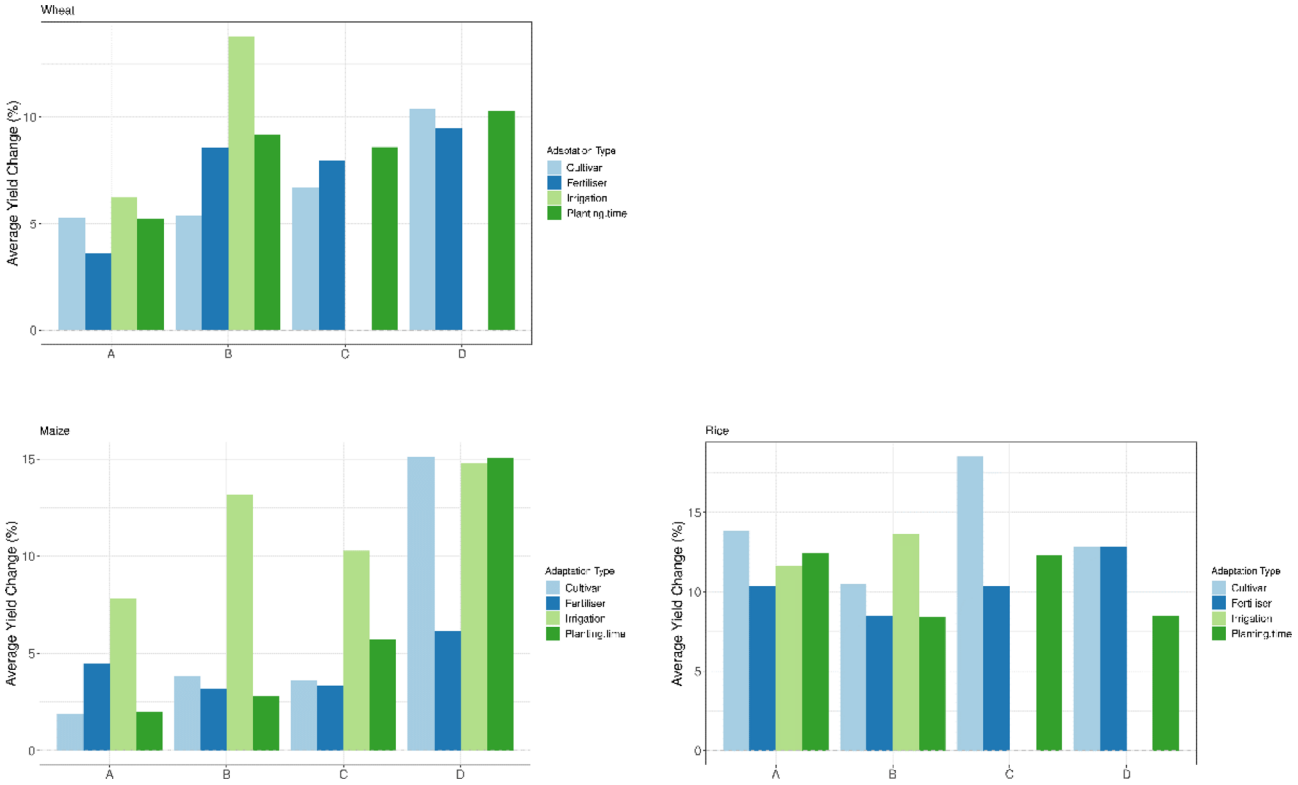


Fig. 3. Yield impacts (%) by distinct adaptation options.

To estimate the threshold effects of climate change on crops, a two-stage approach is applied. In the first stage, Hansen's threshold regression technique is used to endogenously estimate thresholds of temperature change in the relationship between temperature and yields. In the second stage, dataset for each crop is split based on the proposed temperature thresholds and linear mixed-effects models (LMM) are employed to capture the impacts of climate change and adaptation on crop yields in different levels of temperature increase. This approach has three advantages, including interpretability through explicit threshold definition, robust hypothesis testing where the significance of threshold existence is tested via bootstrap methods, and econometric rigor through LMM handling heterogeneity, therefore, appropriate for quantifying threshold effects on global crop yields.

Threshold regression model

Hansen[45]'s threshold regression approach is employed to test whether there exists threshold of temperature change and estimate the threshold values endogenously. The threshold equation that allows the projected yield changes to vary with critical values of temperature change is expressed as follows:

$$RCY_i = \mu + \delta_1 \Delta T_i I(\Delta T \leq \gamma) + \delta_2 \Delta T_i I(\Delta T > \gamma) + \beta_1 \Delta P_i + \beta_2 CO_{2i} + \beta_3 Adapt_i + e_i \quad (1)$$

where $I(\bullet)$ is the indicator function, i denotes the ith observation $(1 \leq i \leq n)$, RCY represents relative change in yields from the baseline period of 2005 to the future-mid point (%). ΔT represents local temperature change from 2005 to the future-mid point (°C) and is the threshold variable, with γ is the threshold value. ΔP is local precipitation change from 2005 (%), CO_2 is mean CO_2 concentration (ppm), and *Adapt* is a dummy variable that equals 1 when at least one adaptation measure is considered in the simulations, and 0 otherwise.

The sum of squared errors of the equation can be minimized by defining OLS estimators for regression parameters $(\widehat{\beta}, \widehat{\delta}, \widehat{\gamma})$, $e_i \sim iidN(0, \sigma^2)$. Once OLS estimators $\widehat{\beta}$ and $\widehat{\delta}$(conditional on γ) are obtained, optimal threshold value γ could be estimated to minimize the concentrated SSE, i.e., $\widehat{\gamma} = \underset{r}{\arg min} SSE_n(\gamma)$.

Because the threshold estimation procedure strictly requires no multi-collinearity among variables, data is preprocessed by eliminating consecutive duplicates in temperature change and precipitation change that appeared across different sites and/or CO_2 concentration scenarios.

Linear mixed-effects model

If a temperature threshold exists for a crop, dataset of that crop is divided into subsets based on temperature change regimes, and linear mixed-effects models are applied to each subset. Using mixed-effects models, rather than directly deriving estimates from a threshold model, allows us to utilize the full set of simulations and incorporate additional variables, such as interactions of climatic factors, thereby improving the assessment of climate impacts on crop yields.

The conventional linear mixed-effects model is represented by the following expression:

$$y_i = X_i\beta + Z_i b_i + \varepsilon_i \tag{2}$$

where y_i is the vector of continuous response, β is the vector of fixed effect parameters for the covariates used in constructing the design matrix X_i. There are two random components: ε_i is the vector of within-group residual errors for group *i* and b_i is the vector of random effect parameters corresponding to random effect design matrix Z_i.

In this study, mixed-effect models with different settings for fixed effects and random effects are tested. In particular, three major climate variables (i.e., temperature change, precipitation change, CO_2 concentration) in mean form, quadratic form, and interaction form (interaction between climate variables) and presence/absence of adaptation measures are tested as fixed effects. Random effects are tested using information of sites (*SiteID*, which combines information of country and publication reference number), climate scenario sources and crop models. Correlation of fixed effects to test for the cross-correlation issue among independent variables. Best-fit models are with no correlation problem detected and with best performance for each crop using two model performance metrics, i.e., coefficient of determination (R^2) and Akaike Information Criterion (AIC). Linear mixed-effects models were fitted to the dataset using the lme4 package in R.

Selected mixed-effects models for wheat by regimes of temperature change:

$$RCY_{L2.38_ij} = \alpha_{01} + \alpha_{11}\Delta T_{ij} + \alpha_{21}CO_{2ij} + \alpha_{31}Adapt_{ij} + \varepsilon_{1ij} \tag{3}$$

$$RCY_{G2.38_ij} = \alpha_{02} + \alpha_{12}\Delta T_{ij} + \alpha_{22}CO_{2ij} + \alpha_{32}Adapt_{ij} + \varepsilon_{2ij} \tag{4}$$

Selected mixed-effects models for wheat by climatic zones:

$$RCY_{A_ij} = \beta_{01} + \beta_{11}\Delta T_{ij} + \beta_{21}\Delta P_{ij} + \beta_{31}\Delta T_{ij}\Delta P_{ij} + \beta_{41}Adapt_{ij} + e_{1ij} \tag{5}$$

$$RCY_{B_ij} = \beta_{02} + \beta_{12}\Delta T_{ij} + \beta_{22}\Delta T_{ij}^2 + \beta_{32}\Delta T_{ij}CO_{2ij} + \beta_{42}Adapt_{ij} + e_{2ij} \tag{6}$$

$$RCY_{C_ij} = \beta_{03} + \beta_{13}\Delta T_{ij} + \beta_{23}\Delta P_{ij} + \beta_{33}CO_{2ij} + \beta_{43}\Delta T_{ij}^2 + \beta_{53}Adapt_{ij} + e_{3ij} \tag{7}$$

$$RCY_{D_ij} = \beta_{04} + \beta_{14}\Delta T_{ij} + \beta_{24}\Delta P_{ij} + \beta_{34}CO_{2ij} + \beta_{44}\Delta T_{ij}^2 + \beta_{54}\Delta T_{ij}\Delta P_{ij} + \beta_{64}Adapt_{ij} + e_{4ij} \tag{8}$$

with.

$$\alpha_0, \beta_0 \sim N\left(\mu_0, \sigma_0^2\right); \alpha_1, \beta_1 \sim N\left(\mu_{\Delta T}, \sigma_{\Delta T}^2\right), and \varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$$

Where *i* and *j* denote simulation at the *jth* row on site *i*. $RCY_{L2.38}$, $RCY_{G2.38}$ are relative changes in wheat yield (%) when temperature increase is less than or greater than the threshold value (e.g., 2.38 °C for wheat), respectively; RCY_A, RCY_B, RCY_C, RCY_D represent relative yield changes in four different climate zones, i.e., tropical zone, arid zone, temperate zone, and continental zone, respectively. ΔT is local temperature change (°C), ΔP is local precipitation change (%), CO_2 is mean CO_2 concentration (ppm), ΔT^2 represents the quadratic term of temperature increase; $\Delta T\Delta P$ and ΔTCO_2 denote interactions between the climatic factors, and *Adapt* represents the presence/absence of adaptation measures. The terms ε and e are within-site residual error random terms. Best-fit LM models for maize and rice are presented in result tables.

Results

Threshold effects of climate change on crop yields

The results of thresholds effects of climate change on crop yields are presented in Table 2. For wheat, there exists a significant single threshold of mean temperature change of 2.38 °C in the relationship between temperature change and relative yield change. When temperature change is lower than 2.38 °C, an increase of 1 °C in temperature would result in a decrease of 6.1% in yields in average over sites. In addition, an increase of 1 ppm in CO_2 concentration would lead to a yield increase of 0.06%. The presence of adaptation practices is found to significantly increase average wheat yield by 6.14%. When average temperature increase exceeds 2.38 °C, the negative impact becomes more severe, as yield loss would increase by 8.2% per °C. In addition, we found a slight decrease in the fertilization effect of CO_2, where a 1 ppm rise in CO_2 concentration would increase wheat yield by 0.056%. Finally, adaptation strategies strongly reduce climate impacts on wheat yields at high temperature regime, where projected wheat yield would increase by 10.3% in response to adaptation practices. The between-site standard deviations of the random parameters are similar before and after the temperature threshold are not higher than the expected values of their coefficients, indicating that temperature effect on yield change of wheat does not vary greatly among sites in each regime of temperature increase.

For rice, threshold of temperature increase is estimated at 3.13 °C. Best-fit models only include temperature change and adaptation availability. It shows that the negative impact of temperature increase is larger at higher regime of temperature, but the positive effect of adaptation also becomes stronger. In particular, when temperature change is less than 3.13 °C, an increase of 1 °C in temperature would result in an average yield loss of 1.07%; while when temperature increase exceeds 3.13 °C, rice yield loss would be 7.14% per °C. The results also show that when adaptation practices are adopted, rice yield would increase by 9 and 20% at low and high regimes of temperature rise, respectively.

Variables	Wheat				Rice				Maize	
	Threshold estimate: $\Delta T^* = 2.38^\circ C$ Bootstrap P–value: 0.00 95% CI: [0.355, 4.913]				Threshold estimate: $\Delta T^* = 3.13^\circ C$ Bootstrap P–value: 0.00 95% CI: [0.355, 4.729]					
	$\Delta T \leq 2.38^\circ C$		$\Delta T > 2.38^\circ C$		$\Delta T \leq 3.13^\circ C$		$\Delta T > 3.13^\circ C$		Estimated coef	Btw–site std. dev
	Estimated coef	Btw–site std. dev	Estimated coef	Btw–site std. dev	Estimated coef	Btw–site std. dev	Estimated coef	Btw–site std. dev		
Intercept	−28.755*** (2.118)	12.599	−19.851*** (3.063)	2.708	−5.025*** (1.022)	7.427	7.086 (11.451)	44.07	0.374 (1.837)	17.893
ΔT	−6.104*** (0.805)	5.316	−8.242*** (1.091)	5.956	−1.066* (0.577)	3.735	−7.14** (3.301)	15.42	−4.031*** (0.535)	4.276
CO_2	0.060*** (0.005)		0.056*** (0.05)						−0.016*** (0.004)	
$\Delta T \Delta P$									0.021*** (0.007)	
Adapt	6.139*** (0.405)		10.283*** (1.193)		9.014*** (0.603)		20.766*** (1.672)		6.223*** (0.583)	
Residual S.D	6.271		10.434		8.050		10.800		10.095	
R^2	0.865		0.860		0.666		0.785		0.821	
No. of obs	1636		576		1205		258		4526	

Table 2. Threshold impacts of climate change on global crop yields. Standard errors in parentheses. ***, **, * denote significance levels of 0.01, 0.05, and 0.1, respectively.

Since the temperature threshold is not determined for maize, the climate impacts on global maize yields are estimated by LM model. The estimation result shows that maize yield would decrease by 4.03% for every 1 °C rise in temperature. The effect of CO_2 on yield is surprisingly negative, but in a small magnitude, with a yield loss of 0.016% per + 1 ppm. The positive and significant interaction between temperature change and precipitation change implies that the negative effect of temperature will be reduced when precipitation increases. Finally, the result confirmed the effectiveness of adaptation strategies where rice yield is projected to increase by 6.2% in response to the presence of adaptation.

Loss in yield for each degree Celsius increase in temperature across thresholds of temperature change are visualized in Fig. 4.

Impacts of climate change on global crop yields by climate zone

Estimated climate impacts and adaptation benefits by distinct climatic zones are displayed in Table 3 (wheat), Table 4 (maize), and Table 5 (rice). The between-site standard deviations of the random coefficients are mostly small compared to their expected values, implying the appropriateness of splitting dataset by major climate zones for examining the projected impacts of climate change on global crop yields.

Wheat

In tropical zones, an increase of 1°C in temperature would lead to a yield loss of 2.61% in wheat. Such negative impact would vary about 2.9% from the mean yield decrease among different sites. Positive effect of precipitation increase on yields is also found, where a 1% increase in precipitation could lead to a yield increase of 0.78%. The negative sign of the temperature-precipitation interaction suggests that an increase in both temperature and precipitation would exacerbate wheat yield loss, likely due to synergistic stress or a diminished benefit of precipitation under higher temperatures. Finally, compared to no adaptation scenarios, the presence of adaptation measures would improve wheat yields by 3.9%.

Wheat yields in arid zones is found to reduce by 8.1% in response to a warming of 1°C, and the negative effect would be amplified if temperature continues to increase, which is reflected by the negative sign of the squared of temperature change. The positive interaction between temperature change and CO_2 implies that the negative impacts of warming on yields is reduced when CO_2 concentration increases. In addition, adaptation strategies would significantly increase wheat yields by 7.5%.

For the temperate (humid subtropical) zone, warmer temperatures reduce wheat yields, but the effect is trivial; however, this negative impact becomes more significant with greater temperature increases, as indicated by the negative sign of the quadratic temperature term. The result also shows that an increase of 1% in precipitation would result in an increase of 0.22% in wheat yield. In addition, wheat yield would increase by 0.05% per + 1 ppm increase in CO_2 concentration. Compared with other climate zones, adaptation strategies are found to has a higher potential for increasing wheat yields in temperate zone (+ 8.05%).

For continental climatic zones, yield loss of wheat would reduce by 2.95% per °C of warming. Similar to the trend found in temperature zones but at larger magnitude, rises in precipitation and CO_2 both found to improve wheat yields in temperature climatic zones, where precipitation increases wheat yields by + 0.67% per 1% and CO_2 concentration improve the yields by + 0.08% per 1 ppm. Also, the positive effect of precipitation could

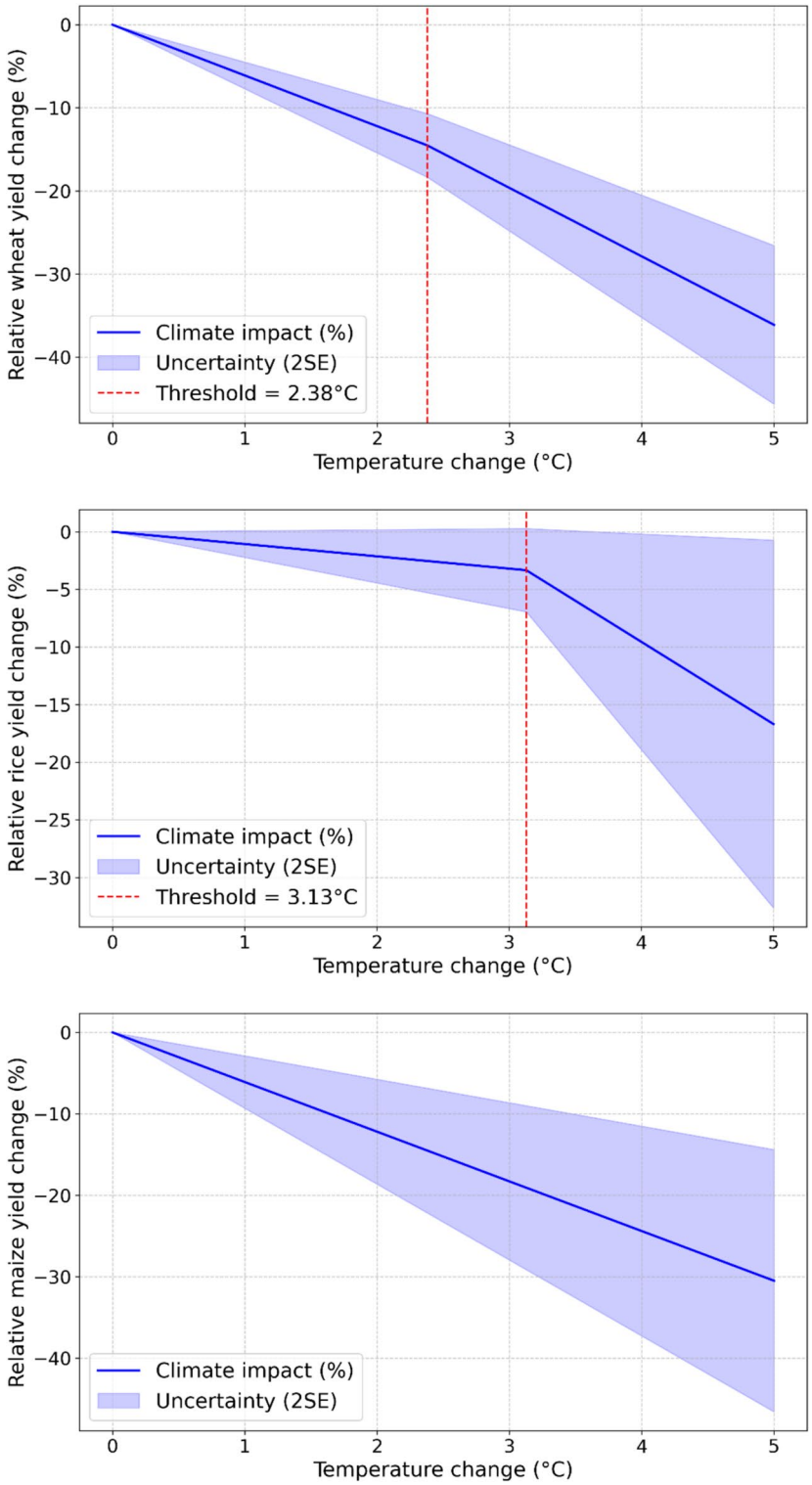


Fig. 4. Estimated climate impacts (%) per 1 °C warming for wheat, rice, and maize, respectively.

be reduced when temperature increases, which is reflected by the negative and significant sign of the $\Delta T \Delta P$ interaction. Finally, the presence of adaptation measures increases wheat yields by + 6.83%.

Maize

The results showed that impact of climate change on maize varies by climate zones. For tropical, arid, temperate, and continental climatic zones, each degree Celsius increase in mean temperature would reduce yields of maize by 1.7%, 4.6%, 3.2%, and 1.8%, respectively. The negative sign of squared of temperature change coefficient for

Variables	A–Tropical climate		B–Arid climate		C–Temperate climate		D–Continental climate	
	Estimated coef	Btw–site std. dev	Estimated coef	Btw–site std. dev	Estimated coef	Btw–site std. dev	Estimated coef	Btw–site std. dev
Intercept	−13.512*** (5.115)	22.768	−7.348*** (1.871)	11.633	−22.067*** (2.867)	10.830	−35.441*** (2.611)	8.821
ΔT	−2.611** (1.260)	2.933	−8.152*** (1.534)	5.351	−2.734 (1.809)	5.062	−2.953* (1.603)	5.024
ΔP	0.787*** (0.261)				0.217** (0.095)		0.667*** (0.165)	
CO_2					0.046*** (0.004)		0.077*** (0.004)	
ΔT^2			−1.318*** (0.319)		−0.737** (0.350)		−0.555** (0.287)	
$\Delta T \Delta P$	−0.325*** (0.120)						−0.216*** (0.065)	
$\Delta T CO_2$			0.017*** (0.001)					
Adapt	3.927*** (0.784)		7.555*** (0.707)		8.054*** (0.864)		6.832*** (0.663)	
Residual S.D	4.329		8.284		5.738		6.762	
R^2	0.969		0.834		0.878		0.855	
No. of obs	163		1055		364		630	

Table 3. Impacts of climate change on wheat yields by climate zones. Standard errors in parentheses. ***, **, * denote significance levels of 0.01, 0.05, and 0.1 respectively. Estimates for wheat in tropical zones are based on limited simulations, while sufficient for our model inclusion, the small sample size may lead to less precise inferences, warranting cautious interpretation.

Variables	A–Tropical climate		B–Arid climate		C–Temperate climate		D–Continental climate	
	Estimated coef	Btw–site std. dev	Estimated coef	Btw–site std. dev	Estimated coef	Btw–site std. dev	Estimated coef	Btw–site std. dev
Intercept	3.663 (3.118)	20.725	−7.528*** (2.584)	18.914	−2.867 (1.988)	9.300	−3.003 (2.045)	11.326
ΔT	−1.709** (0.820)	3.950	−4.571*** (0.927)	4.853	−3.160** (1.599)	4.459	−1.758 (1.423)	3.622
ΔP			0.073** (0.034)				0.391*** (0.136)	
CO_2	−0.040*** (0.005)							
ΔT^2					−0.555* (0.305)		−0.767*** (0.242)	
$\Delta T \Delta P$					0.268*** (0.057)			
Adapt	3.054*** (0.889)		3.803*** (1.393)		6.069*** (1.158)		12.204*** (0.973)	
Residual S.D	7.671		13.842		7.724		8.679	
R^2	0.911		0.713		0.830		0.775	
No. of obs	1392		1249		667		1188	

Table 4. Impacts of climate change on maize yields by climate zones. Standard errors in parentheses, *** and ** denote significance levels of 0.01 and 0.05, respectively.

temperate and continental zones suggested that if temperature continues to rise, the negative impacts of warming would be significantly exacerbated at these regions. Precipitation is also found to improve maize yields in arid and continental zones, with the estimated increase in yields ranges from + 0.07% (arid) to + 0.4% (continental) in response to 1% increase in precipitation. In temperate zones, the precipitation increase would help to reduce the negative effects of warming caused on maize. As an aside, the presence of adaptation practices is projected to significantly improve yields of maize in all four climatic zones, with strong effect found in continental zones (+ 12.2%) and temperate zones (+ 6.1%), followed by dry zones (+ 3.8%), and tropical zones (+ 3.0%).

Rice

Warmer temperatures would create negative impacts on rice yields in regions with dry climate (-3.06% per °C) and temperate climates (−2.18% per °C). In contrast, increased temperature has positive effect on rice yields at high latitudes such as humid continental climates (+ 2.45% per °C). Precipitation increase would improve rice yield in tropical regions, with an increase of 0.67% in rice yields in response to a 1% rise of precipitation level. In tropics, though warming is found to not significantly impact rice production, the negative and significant interaction between temperature change and precipitation change indicates that the positive effect

Variables	A–Tropical climate		B–Arid climate		C–Temperate climate		D–Continental climate	
	Estimated coef	Btw–site std. dev	Estimated coef	Btw–site std. dev	Estimated coef	Btw–site std. dev	Estimated coef	Btw–site std. dev
Intercept	−7.277*** (1.534)	5.104	−5.901* (3.031)	12.731	−3.253 (2.221)	8.325	−1.884 (1.806)	5.016
ΔT	0.370 (0.984)	2.330	−3.058*** (0.881)	3.086	−2.187** (1.098)	4.205	2.454* (1.342)	4.606
ΔP	0.676*** (0.244)							
$\Delta T \Delta P$	−0.456*** (0.116)				−0.554*** (0.09)			
Adapt	10.733*** (0.806)		10.348*** (1.427)		13.498*** (1.444)		7.488*** (1.160)	
Residual S.D	8.853		7.686		8.962		5.104	
R^2	0.538		0.831		0.746		0.885	
No. of obs	712		199		364		138	

Table 5. Impacts of climate change on rice yields by climate zones. Standard errors in parentheses, ***, **, * denote significance levels of 0.01, 0.05, and 0.1, respectively. Estimates for rice in arid and continental zones are based on limited simulations, while sufficient for our model inclusion, these smaller sample sizes may lead to less precise inferences, warranting cautious interpretation.

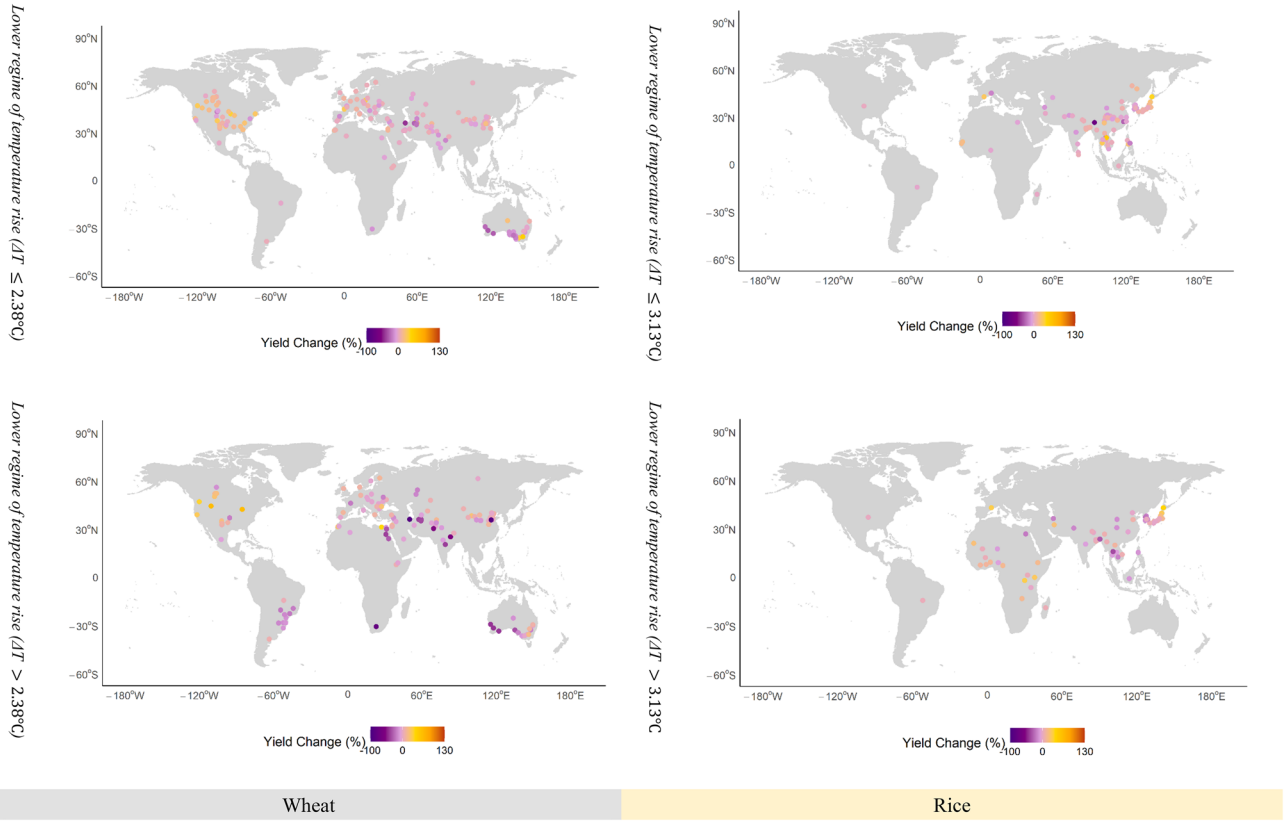


Fig. 5. Climate impacts on crop yields across temperature rise thresholds. Maps are created using ggplot2 3.5.1 (https://github.com/tidyverse/ggplot2/releases/tag/v3.5.1) in R 4.4.3 (https://cran.r-project.org/src/base/R-4).

of precipitation on yield is reduced when temperature increases. Finally, rice demonstrates significant potential for climate adaptation, as the presence of adaptation measures substantially increases crop yields, with yield increases ranging from 7.49% (continental) to 13.49% (temperate).

Spatial variability in projected climate impacts and adaptation benefits by thresholds of temperature rise and climatic zones are visualized on global maps as in Figs. 5, 6, 7.

We also map the global yield change (%) by climatic zones in Fig. 8, where the global map is specified with four different colors representing four major climatic zones (zones are mapped based on the 1-km resolution

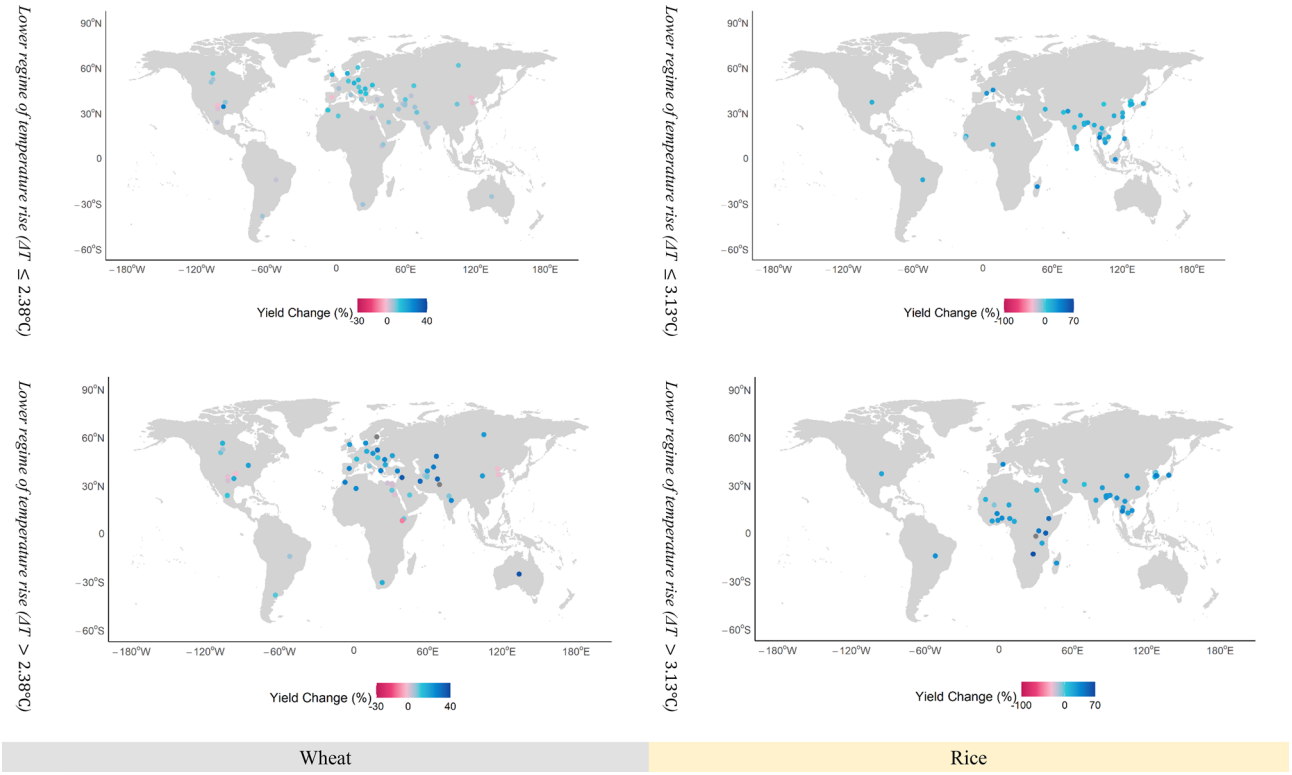


Fig. 6. Adaptation potential across temperature rise thresholds. Maps are created using ggplot2 3.5.1 (https://github.com/tidyverse/ggplot2/releases/tag/v3.5.1) in R 4.4.3 (https://cran.r-project.org/src/base/R-4).

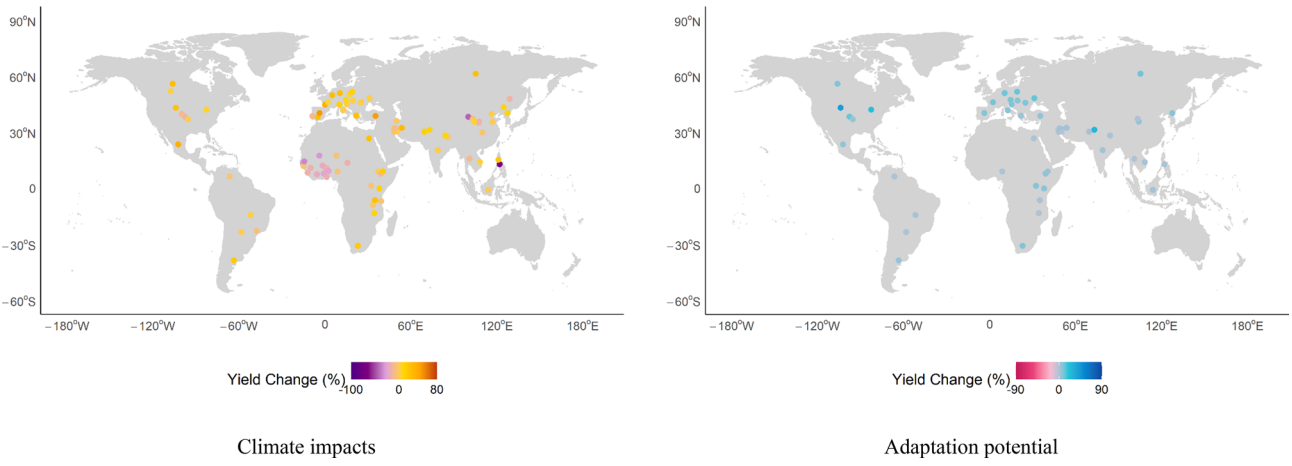


Fig. 7. Climate impacts and adaptation potential on maize yields under RCP8.5 (Mid-Century). Maps are created using ggplot2 3.5.1 (https://github.com/tidyverse/ggplot2/releases/tag/v3.5.1) in R 4.4.3 (https://cran.r-project.org/src/base/R-4).

Koppen 2018 version) and zone color legends provided information of significant estimated yield change per °C of warming.

Spatial variability in adaptation potential (%) are mapped in Fig. 9, where major climatic zones are colorized on the global map, information of significant estimated adaptation benefit is provided on zone color legends.

Suppose that zone-specific effects could stem from differences in baseline temperature and precipitation levels, pooled analyses are performed to assess a broader framework of zone-specific impacts. Besides projected changes (ΔT, ΔP), the 2000–2010 baseline average temperature and precipitation are used as base values in the pooled model. LMMs are applied, selecting best-fit models (based on R^2 and AIC) among all possible combinations of variables, with *base_T* are required to exist in the selected models. The outcomes are presented in Appendix A. For maize, the pooled analysis showed that a 1 °C increase in ΔT will lead to a reduction of

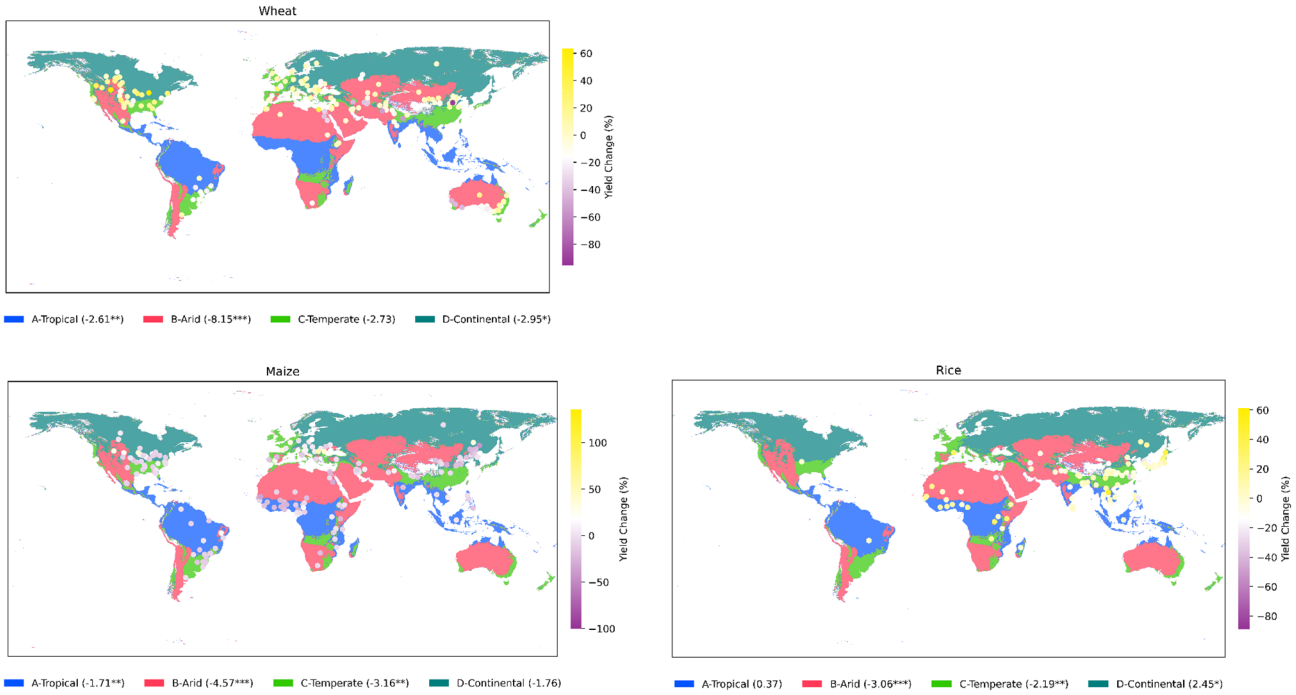


Fig. 8. Climate impacts on global crop yields by climatic zones. Maps created using matplotlib 3.9.3 (https://pypi.org/project/matplotlib/3.9.3) in Python 3.11.3 (https://www.python.org/downloads/release/python-3113/).

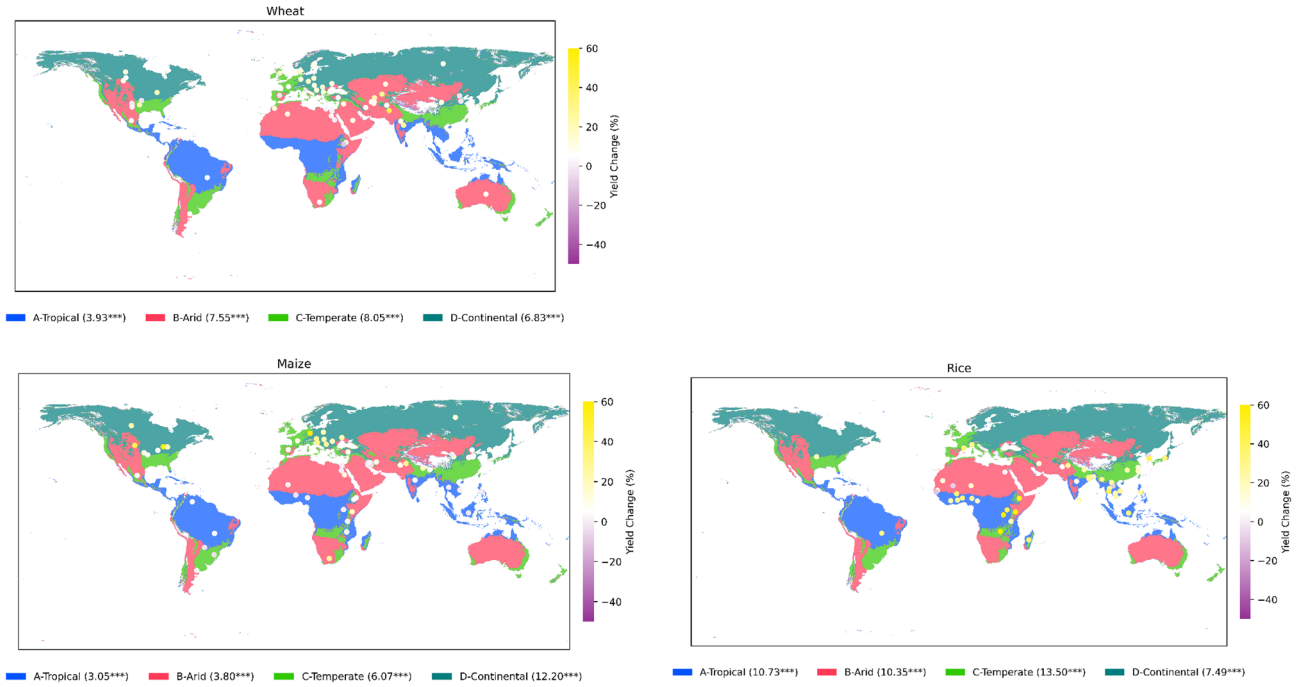


Fig. 9. Adaptation potential for global crop yields by climatic zones. Maps created using matplotlib 3.9.3 (https://pypi.org/project/matplotlib/3.9.3) in Python 3.11.3 (https://www.python.org/downloads/release/python-3113/).

1.69% in yields over sites, aligning closely with those of zones A (−1.7%) and D (−1.8%), but underestimating the larger impacts in zones B (−4.6%) and C (−3.1%). The negative sign of *base_T* coefficient (−0.56%) indicates that hotter baseline temperatures would exacerbate yield losses, contributing −14.25% in Zone A (25.04 °C) versus −5.2% in Zone D (9.27 °C), explaining why hotter zones show more negative ΔT effects. The ten best-fit pooled models do not include *base_P*, while precipitation increase improves maize yields. This result suggested

that though base precipitation does not significantly drive yields, regions with low base level of precipitation (zone B) may increase temperature sensitivity due to water stress, contributing to its large ΔT effect. In general, baseline conditions could partly explain zonal effects, with hotter and drier zones (A, B) facing greater impacts. However, additional factors such as aridity in zone B or underlying interactions in zone C are better captured using zone-specific analysis.

Similar comparison between the two approaches could be made for wheat and rice. Notably, the pooled results for wheat showed a ΔT impact of −7.4% per °C and a *base_T* impact of −0.8% per °C, both are larger than those of maize (ΔT: −1.69%, *base_T*: −0.56%), despite wheat-growing sites having a lower average *base_T* (15 °C) than maize (17.6 °C). This could be explained by wheat's higher sensitivity to temperature. The optimal temperature range of wheat (15–20 °C) is narrower than maize's (20–30 °C), making heat stress, especially around flowering phase, to be harmful to wheat growth and yield. Thus, increase in ΔT and *base_T* both have a more pronounced negative effect on wheat yield compared to maize, even if wheat is grown in cooler regions on average.

Discussion

The threshold values of temperature change in the relationship between climate change and yields of wheat and rice are found at 2.38 and 3.13 °C. The results indicated that temperature rise would reduce crop yields in the first regimes of temperature change (−6.1% to for wheat, −1% for rice) and such negative impacts become more severe in the second regimes (−8.2% for wheat, −7% for rice). The magnitude of impacts is consistent with Zhao, et al.[3]'s findings where global yields of wheat and rice are estimated to reduce by 6.0 and 3.2% in response to 1 °C increase in temperature. The proposed value of thresholds for wheat are also in line with other studies, for example, Xiong, et al.[29] observed a consistent decrease in projected yields of rice, wheat, and maize when mean temperature increase exceeds 2.5 °C. Similarly, Wilcox and Makowski[23] explored the tipping point of 2.3 and 2 °C where more than 50% of simulated relative yield change in wheat are negative when temperature change is higher than 2.3 °C, while Challinor, et al.[21] found steeper declines in wheat yields as temperature increase above 2 °C.

The global mean surface temperature in the baseline period (2001–2010) was approximately 0.79 °C above the pre-industrial period (1850–1900, nominal temperature for 1880)[46]. Thus, the proposed thresholds of temperature rise relative to the pre-industrial period are equivalent to 3.17 °C for wheat and 3.92 °C for rice. These temperature thresholds are quite close to temperature rise under RCP 4.5, RCP 6.0, and RCP8.5 climate change scenarios. Thus, our findings on thresholds of temperature change and the increasing magnitudes of the warming effects on crop yields at higher regimes of temperature confirmed the high risk of crop losses under impacts of climate change associated with medium and high GHG emission scenarios.

In addition, the magnitude of threshold values also indicated that rice is more heat-tolerant than wheat. The finding is appropriate in terms of temperature acclimation of plant growth, where mean optimal temperatures for wheat at different phases of growth (e.g., leaf initiation, shoot growth, root growth, anthesis, and grain filling) are all lower than rice and maize[47]. At different regimes of day/night temperatures, it is found that the maximal rates of photosynthesis are at 25–30 °C for wheat and at 30–35 °C for rice, the net assimilation rate reduced at low temperature for rice but at high temperature for wheat[48].

We found evidences of how the impacts of future climate change on crop yields greatly vary across different climatic zones. First, among four studied climate zones, arid regions are anticipated to the highest crop losses as temperature increase. Though crop yields are found to suffer the largest losses in tropical climates ($\overline{RCY}_{maize}$= −14.5%; $\overline{RCY}_{maize}$= −27.44%, respectively), the projected yield losses per °C increase are found most severe in arid zones, for both wheat and maize. For rice, both average yield loss and crop reduction per 1 °C of warming are projected to be the most in arid zones. This result is differ to previous studies where climate impacts were classified by latitude, in that temperature rise is found to have the most detrimental impacts on crop production in low-latitude regions[40,49,50]. From biophysical science perspectives, higher temperatures mitigate plant maturity and shorten plant development phases, thus causing lower yields in tropical regions rather than in temperate climates. However, crop loss is not only caused by average temperature increases but also by other extreme weather events, especially in tropical climates. In tropics, climate change likely to increase the intensity of these tropical cyclones, such as typhoons and hurricanes. Thus, crop production in tropical regions may suffer greater loss under climate change scenarios but rise in temperature is not always the prominent influencing factor. In addition, mid-to-high latitudes experience a greater increase in temperature and temperature variability than low-latitude regions[51], crops are thus highly sensitive to short-term extreme temperature events and thus experience shorter critical growth periods[52], causing higher latitudes to face more severe crop losses. Also, since insects in tropical zones live with their proximate-to-optimal temperatures and likely to suffer significantly from even minor rise in temperature; while due to cooler baseline climates, warming weathers provide favorable conditions for developments of insects in arid and temperature zones[53,54], leading to a possibility of greater crop reduction in zones B, C due to warming. Moreover, higher temperatures would increase evaporation and worsen water scarcity while baseline water scarcity exacerbates heat stress, both contributing severe crop losses in arid zones. Second, the sign of temperature-precipitation interaction is varying, negative for wheat in tropical and continental zones, for rice in tropical and temperature zones, and positive for maize in temperate zone. While the positive sign is understandable as precipitation would help to reduce the negative impacts caused by warming, the negative interaction sign means that when both temperature and precipitation increase, crop yields would reduce more than expected from individual effects. This could be explained that in crop-growing sites where climatic zones are characterized with high base and project increased precipitation, excessive precipitation could possibly cause waterlogging or worsen heat stress, generating zonal negative $\Delta T \Delta P$ interactions. As an aside, the low R-squared for case of rice in tropical zone possibly stems from the high climate variability and multiple cropping seasons. Compared with wheat and maize, multiple cropping seasons per year for rice in tropical zones

Crop	Author	Datasets and (/or) methods	Yield impacts with 1 °C warming	Uncertainty range
Wheat	Ding, et al.[55]	Field calibration (Luxor Egypt, semi–arid regions)	−17.2% at + 4 °C, or −4.3% per °C[a]	± 1.25%
	Asseng, et al.[31]	Field experiments and ensemble of wheat crop models (global)	−6.0%	± 2.2%
	Zhao, et al.[3]	Multi-methods (global)	−6.0%	± 2.9%
	Wang, et al.[19]	Field experiments (global)	−2.9%	± 2.3%
	Hu, et al.[56]	Multi-methods (global)	−6.0%	± 3.3%
	This study	Crop model simulations (global)	When $\Delta T \leq 2.38^{\circ}$C: −6.1% When $\Delta T > 2.38^{\circ}$C: −8.2%	2SE: ± 1.61%, ± 2.18% SD: ± 5.31%, ± 5.95%
			Tropical zone: −2.61% Arid zone: −8.15% Continental zone: −2.95%	± 2.52% ± 3.07% ± 3.21%
Maize	Zhao, et al.[3]		−7.4%	± 4.5%
	Wang, et al.[19]		−7.1%	± 2.8%
	Hu, et al.[56]		−7.5%	± 5.3%
	This study		−4.03%	± 1.07%
Rice	Zhao, et al.[3]		−3.2%	± 3.7%
	Wang, et al.[19]		−5.6%	± 2.0%
	Hu, et al.[56]		−1.2%	± 5.2%
	This study		When $\Delta T \leq 3.13^{\circ}$C: −1.07% When $\Delta T > 3.13^{\circ}$C: −7.1%	± 1.15% ± 6.6%
Global average crop yields	Lobell and Gourdji[57]	Historical yield data and climate records (global)	~ −5%	−8% to −3%

Table 6. Comparison within existing literature. Uncertainty ranges in previous studies were mostly determined using SD, 2SE, or 95%CI. [a] Linearly scaling the reported 17.2% reduction at + 4 °C, with uncertainty is inferred based on typical crop model ranges ± 5% (Lowell and Burke, 2010).

adding complexity to capturing yield responses, making linear mixed-effects model may struggle to explain. In addition, among four major climatic zones, tropical climate has less distinct seasonal patterns, experiencing higher inter-annual variability with unpredictable frequency of rainfall and extreme weather events, making it trickier to capture yield responses. Thus, future research is recommended to improve the climate impacts assessments for rice in tropical zones.

We compare our findings with those of previous studies in Table 6.

Most of estimated crop responses to warming are consistent with those of previous studies and/or fall within the uncertainty band of their estimates, while a few are different. Besides confirming previous findings, our result could be an extension of existing knowledge by the following points. First, while previous studies treated temperature increases as continuous effect, our study considered thresholds, thus capturing the nuanced effects of climate change on crop yields across different ranges of temperature increase. The significant differences could also partially be explained as refinements within the uncertainty of prior work. In addition, capturing temperature thresholds and zonal effect could improve readers' understanding by connecting methodological approach to actionable insights, highlighting adaptation benefits related to mitigating threshold exceedances or region-specific interventions.

The effect of CO_2 concentration on wheat yields is found to be positive but reduced at higher regimes of temperature change and at hotter and drier climates. For maize, elevated CO_2 causes a slight negative effect on global maize yields in average, and such impact is even more severe in tropical regions. From photosynthesis knowledge, crop yields declined from warmer temperatures could be offset by CO_2 fertilization. However, it is also explored that the compensating CO_2 effects on crops can be offset by heat and drought[38], because high temperatures cause impairment in metabolism and oxidative damage to chloroplasts, thus reducing photosynthesis in plants[58]. As an aside, the photosynthesis rate of C3 crops is found to be more responsive to CO_2 concentration than that of C4 crops[59], explaining the difference in signs of CO_2 effects on wheat and maize.

The quantitative synthesis of the impacts of climate change on global crop yields across thresholds of temperature increase has important policy implications. When temperature increase exceeds its critical high values, warming impacts on crop yields could be turned from positive to negative or from negative to more severely negative. The proposed thresholds of temperature increase, could be locally estimated considering regional practical and socioeconomic constraints, to be better used as critical temperature thresholds for effectively responding to extreme weather events as well as proactively building appropriate long-term adaptation strategies to cope with global warming. As an aside, climate change and adaptation practices can also shift the critical thresholds above which level have a negative effect to higher temperatures. Therefore, further research is recommended to investigate temperature threshold shifts related to yield impacts of climate change considering more related co-occurring drivers, for providing more scientific confidence in adaptation options.

Based on the scientific evidence of the differences in climate impacts and adaptation benefits on crop yields among climatic zones, major producing regions but less affected by climate change should practice intensive crop farming but undertaking sustainably. To increase food production in a low-carbon world, intensive agriculture using more ecological practices would be better than extensive farming over a large area to produce both food and ecosystem services on the same land[60]. In addition, for major crop producing regions under severe impacts

of climate change, strategies and actions for climate adaptation would play a crucial role in combating climate challenges to agricultural production. For example, though many assessments projected severe crop losses under climate change for the 2020s in low-latitude countries, these regions shown large growth in observed yields, and such discrepancy may come from incomplete considerations of technological growth in these studies[61]. For farmers in low-to-middle income countries, major barriers to adapting to climate change include limited access to climate information, technical knowledge, extension services, and financial resources. Thus, it is necessary to break through these challenges before getting onto investments and innovations to facilitate climate adaptation measures.

This study has several limitations. First, the proposed LM models, which treated the aggregated adaptation as a binary variable instead of coding specific adaptation types, may broadly attribute of adaptation effects as the distinct effects of different adaptation interventions are masked, particularly under varying regimes of temperature or precipitation, where, for example, irrigation may mitigate yield losses under heat stress, while cultivar improvement may be more critical under CO_2 fertilization. Second, due to the lack of available data on extreme weather events-such as heatwaves, droughts, or compound extremes-climate impact quantification may be underestimated, especially for maize in arid or tropical regions, which is highly vulnerable to extreme heat during flowering. Third, different process-based crop models vary in their simulations of phenology, CO_2 fertilization, soil processes, or heat stress. As the assessment lacked crop model − specific controls due to data availability issue, the observed variability in crop responses may stem not only from climatological or biological factors but also from methodological differences. Finally, due to limited data, the study could not quantify the interactions between agricultural ecosystem elements − soil properties, water scarcity, pest and disease pressure − with climate factors on crop yields, potentially leading to over- or underestimation of impacts.

Limitations of the present study may guide future research. For instance, future studies should code and model specific adaptation strategies rather than treating aggregated adaptation as a binary variable, enabling a more targeted understanding of which adaptation options are most effective under different temperature and climate zone conditions. Additionally, it is essential to consider extreme heat and rainfall events, accounting for their timing with crop growth stages. Combining gridded extreme event data with yield simulations or empirical models can significantly enhance the realism of impact predictions. For assessments of climate impacts at regional level, future research should examine how different crops interact in mixed or rotational systems, rather than considering monoculture scenarios only, particularly in regions using diversification for resilience. Regarding methodology, a promising approach is combing process-based crop models, remote sensing (RS), and machine learning (ML) to improve spatial prediction accuracy and understand complex, non-linear interactions across scales. For example, Kheir, et al.[62] showed that integrating biophysical models like DSSAT or APSIM with RS vegetation indices and ML algorithms enhances forecasting and interpretation, accounting for dynamic land surface conditions, better capturing feedbacks, and reducing bias in single-model approach.

Conclusion

In this study, we utilize the latest global dataset of site-level process-model crop simulations to quantify the projected impacts of climate change on yields of maize, wheat, and rice, with a special focus on temperature thresholds and climatic zones. To estimate the threshold effects of temperature increases on crop yields, we first employ a threshold regression technique to identify and estimate temperature rise thresholds. The data are then divided into distinct subsets based on these thresholds, and linear mixed-effects models are used to assess climate impacts and adaptation benefits across temperature levels. No significant thresholds were found in the relationship between temperature increases and maize yields; therefore, we estimated average impacts instead and found that maize yields would decrease by 4.03% across countries for a 1 °C warming. For wheat and rice, we identified temperature increase thresholds of 2.38 and 3.13 °C, respectively, with negative impacts becoming more severe when temperature rises exceed these values. For example, our results indicate that a 1 °C increase in temperature would reduce wheat yields by 6.1% when the temperature rise is below 2.38 °C; however, when it exceeds 2.38 °C, the yield loss increases to 8.2% per °C. Similarly, for rice, when the temperature rise surpasses 3.13 °C, the yield reduction worsens from 1.1 to 7.1% per °C increase. These temperature rise thresholds could inform crop-specific emission mitigation targets and be adjusted across local growing sites to develop more tailored adaptation strategies.

To investigate differences in crop responses to climate change across agro-ecological zones, we classified the studied regions by climate zone using the Köppen climate classification system. We found that the negative impacts of warming are most severe in regions with dry climates (arid zones) for all three crops. In contrast, adaptation practices yield positive effects across all climate zones for maize, wheat, and rice, with particularly high adaptive potential for wheat and rice in temperate zones and for maize in continental zones. These findings highlight regional disparities in crop responses and adaptation benefits to global warming, informing targeted policies and interventions to enhance future food security. We also recommend future research to refine climate impact assessment approaches by addressing the limitations discussed earlier in this study.

Data availability

The raw data that support the findings of this study are published by Hasegawa et al.[24], which are available in the figshare repository, https://doi.org/https://doi.org/10.6084/m9.figshare.14691579.v4.

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Author contributions

Bao-Linh Tran conducted data collection and empirical analysis, and wrote the original draft of the manuscript. Wei-Chun Tseng analyzed the methodology and empirical outcomes. Chi-Chung Chen (corresponding author) conceptualized the study, analyzed the empirical results, and suggested policy implications, he also contributed to writing the paper. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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